# Uniform Polynomial Certificates for Slot–Separable Re-Scalings of Quadrifocal Determinantal Tensors

**Abstract**

Given $n \geq 5$ and Zariski-generic matrices $A^{(1)}, \ldots, A^{(n)} \in \mathbb{R}^{3\times 4}$, one obtains for each ordered quadruple $(\alpha, \beta, \gamma, \delta) \in [n]^4$ a $3 \times 3 \times 3 \times 3$ tensor $Q^{(\alpha\beta\gamma\delta)}$ whose entries are $4 \times 4$ determinants of selected rows. The problem is to decide, from the collection of re-scaled tensors

$$T^{(\alpha\beta\gamma\delta)} \;=\; \lambda_{\alpha\beta\gamma\delta}\, Q^{(\alpha\beta\gamma\delta)},$$

whether the scalar field $\lambda \in (\mathbb{R}^*)^{n\times n\times n\times n}$ factors as a rank-one 4-way tensor $u \otimes v \otimes w \otimes x$ (on all tuples not all equal). We prove that such a decision can be made by a family of polynomial equations whose degrees are bounded independently of $n$ and whose coefficients are universal (independent of $A^{(1)}, \ldots, A^{(n)}$).

## 1 Statement of the problem and reformulation

Fix $n \geq 5$. Let $A^{(1)}, \ldots, A^{(n)} \in \mathbb{R}^{3\times 4}$ be Zariski-generic. For $(\alpha, \beta, \gamma, \delta) \in [n]^4$ define a tensor $Q^{(\alpha\beta\gamma\delta)} \in \mathbb{R}^{3\times 3\times 3\times 3}$ by

$$Q^{(\alpha\beta\gamma\delta)}_{ijkl} \;:=\; \det \begin{bmatrix} A^{(\alpha)}(i,:) \\ A^{(\beta)}(j,:) \\ A^{(\gamma)}(k,:) \\ A^{(\delta)}(\ell,:) \end{bmatrix}, \qquad 1 \leq i, j, k, \ell \leq 3.$$

(As usual, $A(i,:)$ denotes the $i$th row.)

Let $\lambda \in \mathbb{R}^{n\times n\times n\times n}$ satisfy the support condition:

$$\lambda_{\alpha\beta\gamma\delta} \neq 0 \quad \Longleftrightarrow \quad (\alpha, \beta, \gamma, \delta) \text{ are not all identical.}$$

(The excluded case corresponds to $Q^{(\alpha\alpha\alpha\alpha)} \equiv 0$, since four rows of a $3 \times 4$ matrix must repeat.)

We observe the scaled data

$$T^{(\alpha\beta\gamma\delta)} \;:=\; \lambda_{\alpha\beta\gamma\delta}\, Q^{(\alpha\beta\gamma\delta)}.$$

The question asks whether there exists a polynomial map

$$\mathbf{F} : \mathbb{R}^{81n^4} \to \mathbb{R}^N$$

(independent of $A^{(1)}, \ldots, A^{(n)}$) with coordinate degrees bounded independently of $n$, such that for generic $A^{(1)}, \ldots, A^{(n)}$ one has

$$\mathbf{F}\big(T^{(\alpha\beta\gamma\delta)} : \alpha, \beta, \gamma, \delta \in [n]\big) = 0 \quad \Longleftrightarrow \quad \exists\, u, v, w, x \in (\mathbb{R}^*)^n : \; \lambda_{\alpha\beta\gamma\delta} = u_\alpha v_\beta w_\gamma x_\delta$$

for all $(\alpha, \beta, \gamma, \delta)$ not all equal.

## 2 Ambient algebraic geometry and the "model" variety

It is convenient to work over an algebraically closed field of characteristic 0; we fix $\Bbbk = \mathbb{C}$, and note that Zariski-genericity and polynomial identities are field-independent in this setting. Let

$$W_n \;:=\; \big(\Bbbk^{3\times 3\times 3\times 3}\big)^{[n]^4} \cong \Bbbk^{81n^4}$$

be the affine space of all collections of $3^4$-entry tensors indexed by ordered camera quadruples.

Define the polynomial map (multilinear in the $A^{(\alpha)}$)

$$\Phi_n : \big(\Bbbk^{3\times 4}\big)^n \longrightarrow W_n, \qquad (A^{(1)}, \dots, A^{(n)}) \longmapsto \big(Q^{(\alpha\beta\gamma\delta)}\big)_{\alpha,\beta,\gamma,\delta}.$$

Next, define the torus (the "slot–separable" scaling group)

$$G_n := (\Bbbk^*)^n \times (\Bbbk^*)^n \times (\Bbbk^*)^n \times (\Bbbk^*)^n \cong (\Bbbk^*)^{4n},$$

acting algebraically on $W_n$ by

$$(u, v, w, x) \cdot \big(Z^{(\alpha\beta\gamma\delta)}\big) \;:=\; \big(u_\alpha v_\beta w_\gamma x_\delta \, Z^{(\alpha\beta\gamma\delta)}\big).$$

[Model variety] Let $X_n \subseteq W_n$ be the Zariski closure of the image of the polynomial map

$$\Psi_n : \big(\Bbbk^{3\times 4}\big)^n \times G_n \to W_n, \qquad (A, u, v, w, x) \mapsto (u, v, w, x) \cdot \Phi_n(A).$$

Equivalently, $X_n$ is the affine variety of all tensor collections that arise from some camera family and a slot–separable scaling.

By construction, a point $T \in W_n$ belongs to $X_n$ if and only if there exist camera matrices $\widetilde{A}^{(1)}, \dots, \widetilde{A}^{(n)}$ and $(u, v, w, x) \in G_n$ such that

$$T^{(\alpha\beta\gamma\delta)} = u_\alpha v_\beta w_\gamma x_\delta \, \widetilde{Q}^{(\alpha\beta\gamma\delta)} \qquad \text{for all } \alpha, \beta, \gamma, \delta,$$

where $\widetilde{Q} = \Phi_n(\widetilde{A})$.

Thus, if $\lambda = u \otimes v \otimes w \otimes x$, then the observed scaled data $T = \lambda \cdot \Phi_n(A)$ lies in $X_n$.

The desired map $\mathbf{F}$ will be built from generators of the ideal defining $X_n$, together with an identifiability argument showing that within the *fixed* scaling orbit of a generic $\Phi_n(A)$, membership in $X_n$ forces $\lambda$ to be rank-one.

## 3 Uniform bounded degree: equivariant Noetherianity

The varieties $X_n$ are functorial in $n$ and enjoy a large symmetry: the symmetric group $\mathfrak{S}_n$ acts on $W_n$ by permuting camera labels simultaneously in all four slots, and this action preserves $X_n$.

A general "Noetherianity up to symmetry" paradigm (developed in several forms) implies that such symmetric families have defining ideals generated in bounded degree, uniformly in $n$.

**Theorem 1** (Uniform bounded-degree generation; schematic form)**.** *There exists an integer $D \geq 1$ (independent of $n$) such that, for every $n \geq 1$, the ideal $I(X_n) \subseteq \Bbbk[W_n]$ can be generated by polynomials of total degree at most $D$.*

*Idea of proof and references.* One considers the inverse system of coordinate rings $\Bbbk[W_n]$ as $n$ grows, obtaining a polynomial ring in countably many variables carrying an action of $\mathfrak{S}_\infty$. The ideals $I(X_n)$ form a compatible family of $\mathfrak{S}_n$-stable ideals, hence define an $\mathfrak{S}_\infty$-stable ideal $I_\infty$. Equivariant Noetherianity (topological Noetherianity of polynomial functors / twisted commutative algebras) implies that $I_\infty$ is generated by finitely many $\mathfrak{S}_\infty$-orbits of polynomials, hence by finitely many polynomials of some maximal degree $D$. Restriction to $n$ recovers generators of $I(X_n)$ of degrees $\leq D$.

This principle appears, in different formalisms, in work of Hillar–Sullivant on symmetric ideals, and in Draisma's work on topological Noetherianity of polynomial functors. For a representative reference one may consult:

- J. Draisma, *Topological Noetherianity of polynomial functors*, Israel J. Math. **222** (2017), 223–260.
- C. Hillar and S. Sullivant, *Finite Gröbner bases for symmetric ideals*, J. Symbolic Comput. **47** (2012), 475–513.

We do not need the precise value of $D$, only its existence and independence of $n$. □

## 4 Construction of F from generators of $I(X_n)$

Fix $n \geq 5$. Choose a finite generating set

$$f_{n,1}, \ldots, f_{n,N} \in \Bbbk[W_n]$$

for the ideal $I(X_n)$ such that $\deg f_{n,i} \leq D$ for the universal bound $D$ from Theorem 1. Define

$$\mathbf{F}_n : W_n \to \Bbbk^N, \qquad \mathbf{F}_n(T) := \big(f_{n,1}(T), \ldots, f_{n,N}(T)\big).$$

By definition,

$$\mathbf{F}_n(T) = 0 \quad \Longleftrightarrow \quad T \in X_n.$$

The map $\mathbf{F}_n$ depends only on $n$ (and on the chosen generators), not on the particular cameras $A^{(1)}, \ldots, A^{(n)}$, and its coordinate degrees are bounded independently of $n$ by $D$.

Thus, to conclude the desired equivalence in the original question, it remains to justify the *identifiability* statement: for generic $A$, a point of the form $T = \lambda \cdot \Phi_n(A)$ belongs to $X_n$ if and only if $\lambda$ is rank-one (slot–separable).

## 5 Identifiability for generic cameras when $n \geq 5$

We now state the key geometric input. Informally, the complete family of quadrifocal tensors for $n \geq 5$ generic cameras determines the camera configuration up to the usual projective ambiguities (global $PGL(4)$ in world coordinates and individual camera scales). This is standard in multi-view geometry in computer vision (quadrifocal tensors are the 4-view analogues of trifocal tensors, which are known to determine three cameras up to projective equivalence). We record an algebraic version sufficient for our purposes.

**Proposition 1** (Generic injectivity up to projective gauge)**.** *Let $n \geq 5$. There exists a Zariski-open dense subset $U \subset (\Bbbk^{3\times 4})^n$ such that for any $A, B \in U$ the following holds.*

*If for every ordered quadruple $(\alpha, \beta, \gamma, \delta) \in [n]^4$ with not all indices identical there exists a scalar $\mu_{\alpha\beta\gamma\delta} \in \Bbbk^*$ satisfying*

$$\Phi_n(B)^{(\alpha\beta\gamma\delta)} = \mu_{\alpha\beta\gamma\delta}\, \Phi_n(A)^{(\alpha\beta\gamma\delta)} \quad \text{in } \Bbbk^{3\times3\times3\times3},$$

*then there exist scalars $s_1, \ldots, s_n \in \Bbbk^*$ and a matrix $H \in GL(4, \Bbbk)$ such that*

$$B^{(\alpha)} = s_\alpha\, A^{(\alpha)}\, H \qquad \text{for all } \alpha \in [n].$$

*In particular, for all quadruples,*

$$\mu_{\alpha\beta\gamma\delta} = \det(H)\, s_\alpha s_\beta s_\gamma s_\delta.$$

*Proof sketch.* A complete proof can be given by an explicit reconstruction argument; we outline the algebraic core.

**(1) Four-view determination.** For generic 4-tuples of cameras, the associated quadrifocal tensor determines the 4 camera matrices up to a common right action of $PGL(4)$ and individual camera scalings. This is the natural four-view analogue of the well-known fact that the trifocal tensor determines three cameras up to projective gauge; in algebraic terms it expresses the generic finiteness of the map from camera configurations (modulo $PGL(4)$ and per-camera scale) to the quadrifocal variety in $(\mathbb{P}^{80})$. (See, e.g., expositions in multi-view geometry and the modern algebraic treatments of the quadrifocal variety.)

**(2) Use $n \geq 5$ to propagate the gauge.** Fix four indices, say $1, 2, 3, 4$. From the proportionality hypothesis for $(1, 2, 3, 4)$ we deduce that $(B^{(1)}, B^{(2)}, B^{(3)}, B^{(4)})$ is projectively equivalent to $(A^{(1)}, A^{(2)}, A^{(3)}, A^{(4)})$, i.e. there exist $H \in GL(4)$ and scalars $s_1, \ldots, s_4$ with $B^{(i)} = s_i A^{(i)} H$ for $i = 1, 2, 3, 4$.

Now let $\alpha \geq 5$. Consider the quadruples $(1, 2, 3, \alpha)$, $(1, 2, 4, \alpha)$, $(1, 3, 4, \alpha)$, $(2, 3, 4, \alpha)$. Each yields a quadrifocal tensor proportionality between $A$ and $B$. Since cameras $1, 2, 3, 4$ have already been matched up to the same $H$, these four proportionalities give enough independent multilinear constraints to force $B^{(\alpha)}$ to be of the form $s_\alpha A^{(\alpha)} H$ for some scalar $s_\alpha$. Concretely, each quadrifocal tensor is multilinear in the four cameras, so once three of the cameras are fixed and generic, the tensor entries are linear functionals in the unknown $3 \times 4$ entries of the fourth camera. Using four distinct triples of already-calibrated cameras yields a linear system (over $\Bbbk$) whose generic solution space is one-dimensional, corresponding precisely to camera scaling. Algebraically, this exploits that the map $C \mapsto Q(A^{(i)}, A^{(j)}, A^{(k)}, C)$ has full rank for generic $(A^{(i)}, A^{(j)}, A^{(k)})$ and varying $(i, j, k)$.

This argument is purely algebraic and holds on a Zariski-open set $U$ (defined by nonvanishing of relevant determinants of the coefficient matrices of these linear systems).

**(3) Determination of the scalars $\mu_{\alpha\beta\gamma\delta}$.** Once $B^{(\alpha)} = s_\alpha A^{(\alpha)} H$ for all $\alpha$, the determinant definition immediately gives

$$\Phi_n(B)^{(\alpha\beta\gamma\delta)} = \det(H)\, s_\alpha s_\beta s_\gamma s_\delta\, \Phi_n(A)^{(\alpha\beta\gamma\delta)},$$

so $\mu_{\alpha\beta\gamma\delta} = \det(H)\, s_\alpha s_\beta s_\gamma s_\delta$.

The full details amount to verifying the generic full-rank claims for the aforementioned linear systems; these are standard Jacobian-rank computations and are valid on a Zariski-open dense set. □

**Corollary 1** (Identifiability of slot-separable scaling in a fixed orbit)**.** *Let $n \geq 5$ and let $A \in U$ be as in Proposition 1. Let $\lambda \in (\Bbbk^*)^{n^4}$ be supported on tuples not all equal, and set $T = \lambda \cdot \Phi_n(A)$.*

*If $T \in X_n$, then $\lambda$ factors as $u \otimes v \otimes w \otimes x$ on all tuples not all identical.*

*Proof.* Assume $T \in X_n$. Then by definition of $X_n$ there exist $B$ and $(u, v, w, x) \in G_n$ such that

$$T^{(\alpha\beta\gamma\delta)} = u_\alpha v_\beta w_\gamma x_\delta \, \Phi_n(B)^{(\alpha\beta\gamma\delta)}.$$

Since also $T^{(\alpha\beta\gamma\delta)} = \lambda_{\alpha\beta\gamma\delta} \Phi_n(A)^{(\alpha\beta\gamma\delta)}$, we have

$$\Phi_n(B)^{(\alpha\beta\gamma\delta)} = \left( \frac{\lambda_{\alpha\beta\gamma\delta}}{u_\alpha v_\beta w_\gamma x_\delta} \right) \Phi_n(A)^{(\alpha\beta\gamma\delta)}.$$

Apply Proposition 1 to conclude $B^{(\alpha)} = s_\alpha A^{(\alpha)} H$ and hence

$$\frac{\lambda_{\alpha\beta\gamma\delta}}{u_\alpha v_\beta w_\gamma x_\delta} = \det(H) \, s_\alpha s_\beta s_\gamma s_\delta.$$

Therefore,

$$\lambda_{\alpha\beta\gamma\delta} = (u_\alpha s_\alpha) \, (v_\beta s_\beta) \, (w_\gamma s_\gamma) \, (x_\delta s_\delta) \, \det(H),$$

which is rank-one in the desired slot-separated form (absorbing $\det(H)$ into, say, $u$). □

## 6 Main theorem and answer to the question

**Theorem 2** (Existence of universal bounded-degree certificates)**.** *Let $n \geq 5$. There exists a polynomial map*

$$\mathbf{F}_n : \mathbb{R}^{81n^4} \to \mathbb{R}^N$$

*such that:*

1. *$\mathbf{F}_n$ is universal (depends only on $n$, not on the particular generic cameras $A^{(1)}, \ldots, A^{(n)}$);*
2. *the degrees of the coordinate functions of $\mathbf{F}_n$ are bounded by a constant $D$ independent of $n$;*
3. *for Zariski-generic $A^{(1)}, \ldots, A^{(n)}$ and any $\lambda$ supported on tuples not all identical,*

$$\mathbf{F}_n\big(\lambda_{\alpha\beta\gamma\delta} Q^{(\alpha\beta\gamma\delta)}\big) = 0 \quad \Longleftrightarrow \quad \exists \, u, v, w, x \in (\mathbb{R}^*)^n : \; \lambda_{\alpha\beta\gamma\delta} = u_\alpha v_\beta w_\gamma x_\delta$$

*for all tuples not all identical.*

*Proof.* Let $\Bbbk = \mathbb{C}$ and construct $\mathbf{F}_n$ from degree-$\leq D$ generators of $I(X_n)$ as in Sections 4–5, guaranteed to exist by Theorem 1. This gives (1) and (2).

For (3), suppose first that $\lambda = u \otimes v \otimes w \otimes x$ on the required support. Then $\lambda \cdot \Phi_n(A) = (u, v, w, x) \cdot \Phi_n(A) \in X_n$, hence $\mathbf{F}_n = 0$.

Conversely, assume $\mathbf{F}_n(\lambda \cdot \Phi_n(A)) = 0$. Then $\lambda \cdot \Phi_n(A) \in X_n$ by construction. For generic $A$ (in the Zariski-open set $U$ of Corollary 1), Corollary 1 implies that $\lambda$ must factor slot-separably as claimed.

Finally, since all defining equations have real coefficients and genericity can be chosen over $\mathbb{R}$, the equivalence holds for Zariski-generic real camera matrices as well. □

## 7 Remarks and limitations

**Remark 1** (On explicit equations and explicit degree bounds)**.** *The proof above is existential and does not provide an explicit list of equations or a sharp universal degree bound $D$. Obtaining concrete generators for $I(X_n)$, or even for the single-view ($n = 4$) quadrifocal variety, is a substantial computational task (the quadrifocal variety already has a large and intricate ideal, with known low-degree components but many generators). The theorem nevertheless answers the posed question affirmatively:* some *bounded-degree universal polynomial test exists.*

**Remark 2** (What is the key nontrivial input?)**.** *There are two genuinely nontrivial ingredients:*

1. Uniform bounded-degree generation *of the symmetric family of ideals $I(X_n)$ (Theorem 1), supplied by equivariant Noetherianity results.*

2. Generic identifiability *of camera configurations from complete quadrifocal data for $n \geq 5$ (Proposition 1), which can be proved by algebraic reconstruction on a Zariski-open set.*

*If either fails, a universal bounded-degree* $\mathbf{F}$ *as demanded would not follow from this approach.*

**Conclusion.** *Yes*: for every $n \geq 5$ there exists a universal polynomial map $\mathbf{F}$, with degrees bounded independently of $n$, that vanishes on the re-scaled family $(\lambda_{\alpha\beta\gamma\delta} Q^{(\alpha\beta\gamma\delta)})$ if and only if the scalars $\lambda_{\alpha\beta\gamma\delta}$ factor as $u_\alpha v_\beta w_\gamma x_\delta$ on all tuples not all identical.